

Attendance & Leave Management System

Automated Tracking and Approval Workflow

By Atanu Paul | 14 Apr 2025 | Google Sheet

Introduction

Objective: Streamline attendance tracking and leave management.

Key Features:

- Real-time attendance marking.
- Automated leave approval based on thresholds.
- Managerial override capabilities.
- Integration with billing systems.

Attendance Marking

Options Available:

- Present
- Half Day
- Full Day Leave (PL)
- Half Day Leave (PL Half)
- UPL
- UPL - Half
- Week Off
- Holiday
- Rejected
- Approved
- Approved Half

Mechanism: Users mark attendance via a shared Google Sheet.

Automated Leave Approval

Process:

- On edit, script calculates total leave requests for the day.
- If requests exceed maximum allowed, latest request is marked as "Rejected".
- Email notifications sent to users regarding approval or rejection.

Dear ~~XXXXXXXXXX~~

Your leave request for Tue Apr 01 2025 has been approved.

Please note, this is an auto-generated mail. Your leave approval will be officially reflected in the sheet after confirmation from L1. While your leave has been approved, it is subject to cancellation at any time due to business needs.

Best Regards,
~~XXXXXXXXXX~~

Dear ~~XXXXXXXXXX~~

Your leave request for Tue Feb 11 2025 only one leave is allowed on that date, and it has already been approved. The request has been marked as "Rejected". Please contact your team lead or manager for further consideration.

Best Regards,
~~XXXXXXXXXX~~

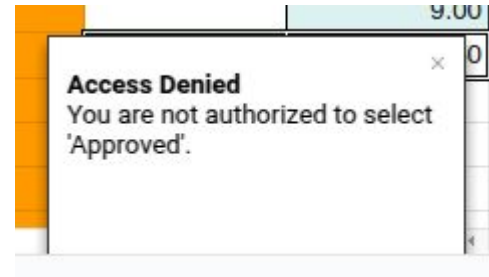
Managerial Overrides

Capability:

- Managers can manually approve or reject leave requests.
- Options: "Approved", "Approved Half".

Access Control:

- Dropdown options restricted to managerial roles.



Email Notifications

Approved Leave:

- Subject: "Leave Approved"
- Body: "Your leave for [Date] has been approved. Subject to change based on business needs."

Rejected Leave:

- Subject: "Leave Request Reconsideration Needed"
- Body: "Your leave request for [Date] has been rejected due to exceeding the maximum allowed leaves for the day."

Shrinkage & Billing Integration

- **Shrinkage Calculation:**
 - System computes daily headcount shrinkage.
- **Billing:**
 - Data integrates with billing systems to adjust charges based on attendance.

TOTAL LEAVES(PLANNED & UNPLANNED)	
SUM of Planned & Unplanned Leaves	18.50
Team's Shrinkage %	3.82%

Security & Access Control

- **User Roles:**
 - Team Members: Can mark attendance and apply for leave.
 - Managers: Can approve/reject leave requests.
- **Data Integrity:**
 - Scripts ensure only authorized edits are accepted.

Sample Sheet

April - 2025									AUGUST ATTENDANCE SUMMARY(BILLED DAYS)							
EMP. ID.	NAME	Email	Center		Thu	Fri	Sat	Sun								
					10-Apr	11-Apr	12-Apr	13-Apr	#PRESENT	#UNPLANNED LEAVE	#UNPLANNED LEAVE %	#PLANNED LEAVE	#PLANNED LEAVE %	#ATTENDANCE %	#SHRINKAGE %	TOTAL WORKING DAYS
				Approved Half Day	Approved	WO	WO		7.00	0.00	0.00%	6.50	29.55%	31.82%	29.55%	22
				Present	Present	WO	WO		9.00	0.00	0.00%	0.00	0.00%	40.91%	0.00%	22
				Present	Present	WO	WO		9.00	0.00	0.00%	0.00	0.00%	40.91%	0.00%	22
				Present	Present	WO	WO		9.00	0.00	0.00%	1.50	6.82%	40.91%	6.82%	22
				Present	Present	WO	WO		10.00	0.00	0.00%	2.00	9.09%	45.45%	9.09%	22
				Present	Present	WO	WO		9.50	0.00	0.00%	0.50	2.27%	43.18%	2.27%	22
				Present	Present	WO	WO		9.00	0.00	0.00%	1.00	4.55%	40.91%	4.55%	22
				Present	Present	WO	WO		10.00	0.00	0.00%	0.00	0.00%	45.45%	0.00%	22
				Present	Present	WO	WO		10.00	0.00	0.00%	0.00	0.00%	45.45%	0.00%	22
				Present	Present	WO	WO		10.00	0.00	0.00%	0.00	0.00%	45.45%	0.00%	22
				Present	Present	WO	WO		10.00	0.00	0.00%	0.00	0.00%	45.45%	0.00%	22
				Present	Present	WO	WO		9.00	0.00	0.00%	0.00	0.00%	40.91%	0.00%	22
				Present	PL	WO	WO		9.00	0.00	0.00%	3.00	13.64%	40.91%	13.64%	22
				Present	Present	WO	WO		10.00	0.00	0.00%	0.00	0.00%	45.45%	0.00%	22
				Present	Present	WO	WO		10.00	0.00	0.00%	0.00	0.00%	45.45%	0.00%	22
				Present	Present	WO	WO		10.00	0.00	0.00%	0.00	0.00%	45.45%	0.00%	22
				TOTAL PRESENT					21	20	0	0	206.50	0.00	0.00%	18.50
TOTAL UNPLANNED LEAVE					0	0	0	0								
TOTAL PLANNED LEAVE					0	1	0	0								
TOTAL HALF-DAY					0	0	0	0								
DAY-WISE SHRINKAGE				0.00%	0.00%	3.33%	0.00%	0.00%								

Benefits

- **Efficiency:** Reduces manual tracking and approvals.
- **Transparency:** Clear visibility of attendance records.
- **Scalability:** Adaptable to teams of various sizes.
- **Integration:** Seamless connection with billing systems.

Conclusion

- **Summary:** The system offers a comprehensive solution for attendance and leave management, ensuring efficiency and accuracy.